Task 1:

Code:

Terminal

```
#!/bin/bash

awk -F',' 'NR>1 {
  total = $3+$4+$5
  avg=total/3
  if( avg >= 75 )
          grade = "Distinction"
  else if ( avg >= 40 )
          grade = "Pass"
  else
          grade = "Fail"
  printf "Roll No: %d, Name: %s, Total: %d, Average: %.2f, Grade: %s\n", $1 , $2 ,total ,avg , grade }' "$1" >> report_awk.txt

echo "Report Saved"
```

Output:

Terminal

```
bodike@bodike-chaithali:~/Scripts$ ls
bodike@bodike-chaithali:~/Scripts$ vi B24CS013_06_01.sh
bodike@bodike-chaithali:~/Scripts$ ls
B24CS013_06_01.sh
bodike@bodike-chaithali:~/Scripts$ cat B24CS013_06_01.sh
#!/bin/bash

awk -F',' 'NR>1 {
  total = $3+$4+$5
  avg=total/3
  if( avg >= 75 )
          grade = "Distinction"
  else if ( avg >= 40 )
          grade = "Pass"
  else
          grade = "Fail"
  printf "Roll No: %d, Name: %s, Total: %d, Average: %.2f, Grade: %s\n", $1 , $2 ,total ,avg , grade }' "$1" >> report_awk.txt

echo "Report Saved"
bodike@bodike-chaithali:~/Scripts$ cat >> students.csv << EOF
> RollNo,Name,Maths,Science,English
> 101,John,85,78,92
> 102,Alice,45,50,42
> 103,Bob,70,68,72
> 104,Eve,90,88,95
> 105,Tom,30,35,28
> EOF
bodike@bodike-chaithali:~/Scripts$ cat students.csv
RollNo,Name,Maths,Science,English
101,John,85,78,92
102,Alice,45,50,42
103,Bob,70,68,72
104,Eve,90,88,95
105,Tom,30,35,28
bodike@bodike-chaithali:~/Scripts$ ls -l B24CS013_06_01.sh
-rw-rw-r-- 1 bodike bodike 346 Aug 21 16:48 B24CS013_06_01.sh
bodike@bodike-chaithali:~/Scripts$ chmod +x B24CS013_06_01.sh
bodike@bodike-chaithali:~/Scripts$ ls -l B24CS013_06_01.sh
```

```
bodike@bodike-chaithali:~/Scripts$ ls -l B24CS013_06_01.sh
-rw-rw-r-- 1 bodike bodike 346 Aug 21 16:48 B24CS013_06_01.sh
bodike@bodike-chaithali:~/Scripts$ chmod +x B24CS013_06_01.sh
bodike@bodike-chaithali:~/Scripts$ ls -l B24CS013_06_01.sh
-rwxrwxr-x 1 bodike bodike 346 Aug 21 16:48 B24CS013_06_01.sh
bodike@bodike-chaithali:~/Scripts$ ./B24CS013_06_01.sh students.csv
Report Saved
bodike@bodike-chaithali:~/Scripts$ ls
B24CS013_06_01.sh  report_awk.txt  students.csv
bodike@bodike-chaithali:~/Scripts$ cat report_awk.txt
Roll No: 101, Name: John, Total: 255, Average: 85.00, Grade: Distinction
Roll No: 102, Name: Alice, Total: 137, Average: 45.67, Grade: Pass
Roll No: 103, Name: Bob, Total: 210, Average: 70.00, Grade: Pass
Roll No: 104, Name: Eve, Total: 273, Average: 91.00, Grade: Distinction
Roll No: 105, Name: Tom, Total: 93, Average: 31.00, Grade: Fail
bodike@bodike-chaithali:~/Scripts$
```

Task 2:

Code:

Terminal

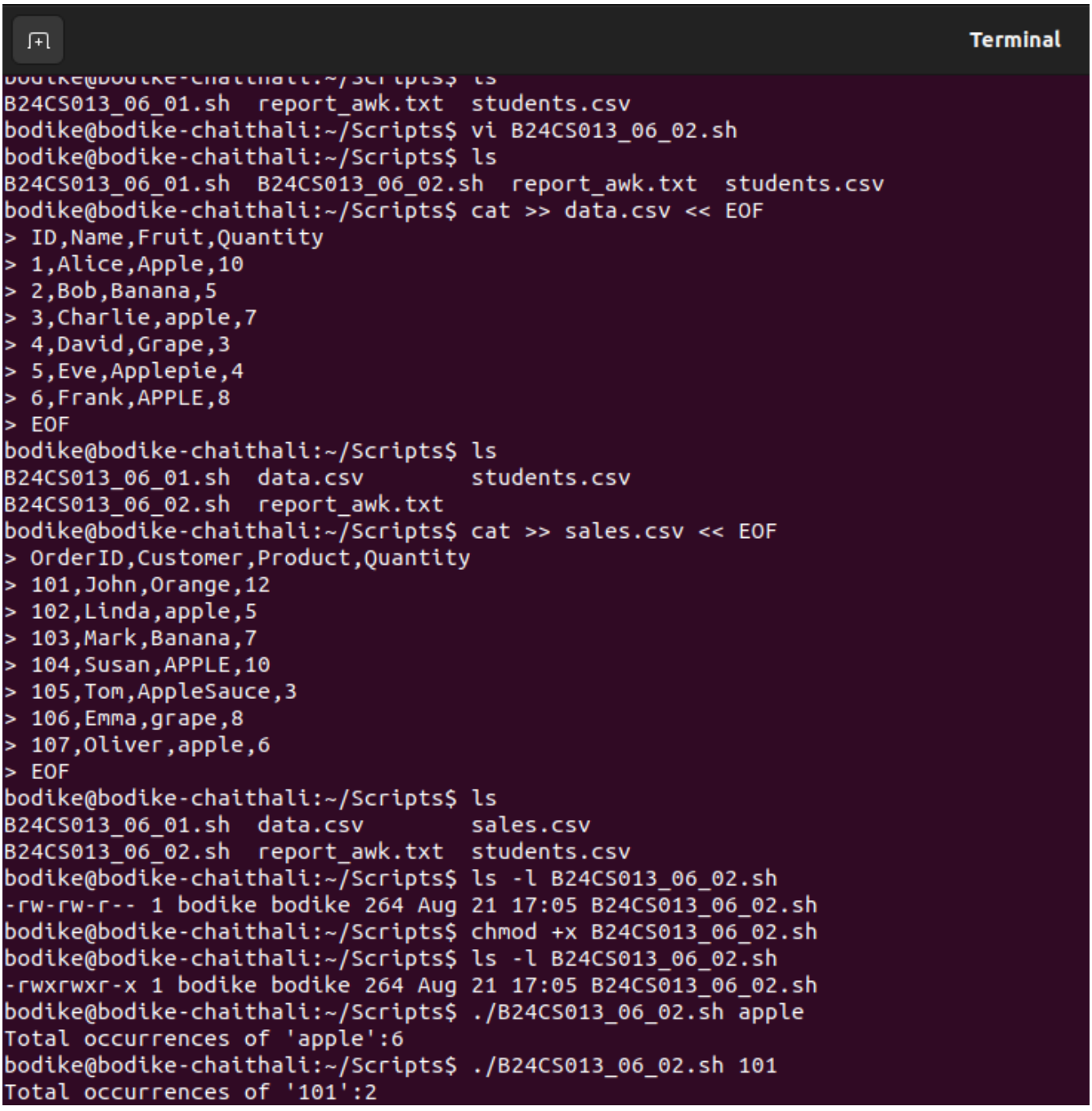
```
#!/bin/bash

if [ $# -ne 1 ]; then
        echo "Usage: $0 word"
        exit 1
fi

word=$1
count=$(grep -iowh "$word" *.csv | wc -l)

if [ $? -ne 0 ]; then
        echo "No .csv files found"
else
        echo "Total occurrences of '$word':$count"
fi
```

Output:


Terminal
B24CS013_06_01.sh report_awk.txt students.csv
bodike@bodike-chaithali:~/Scripts$ vi B24CS013_06_02.sh
bodike@bodike-chaithali:~/Scripts$ ls
B24CS013_06_01.sh B24CS013_06_02.sh report_awk.txt students.csv
bodike@bodike-chaithali:~/Scripts$ cat >> data.csv << EOF
> ID,Name,Fruit,Quantity
> 1,Alice,Apple,10
> 2,Bob,Banana,5
> 3,Charlie,apple,7
> 4,David,Grape,3
> 5,Eve,Applepie,4
> 6,Frank,APPLE,8
> EOF
bodike@bodike-chaithali:~/Scripts$ ls
B24CS013_06_01.sh data.csv   students.csv
B24CS013_06_02.sh report_awk.txt
bodike@bodike-chaithali:~/Scripts$ cat >> sales.csv << EOF
> OrderID,Customer,Product,Quantity
> 101,John,Orange,12
> 102,Linda,apple,5
> 103,Mark,Banana,7
> 104,Susan,APPLE,10
> 105,Tom,AppleSauce,3
> 106,Emma,grape,8
> 107,Oliver,apple,6
> EOF
bodike@bodike-chaithali:~/Scripts$ ls
B24CS013_06_01.sh data.csv   sales.csv
B24CS013_06_02.sh report_awk.txt students.csv
bodike@bodike-chaithali:~/Scripts$ ls -l B24CS013_06_02.sh
-rw-rw-r-- 1 bodike bodike 264 Aug 21 17:05 B24CS013_06_02.sh
bodike@bodike-chaithali:~/Scripts$ chmod +x B24CS013_06_02.sh
bodike@bodike-chaithali:~/Scripts$ ls -l B24CS013_06_02.sh
-rwxrwxr-x 1 bodike bodike 264 Aug 21 17:05 B24CS013_06_02.sh
bodike@bodike-chaithali:~/Scripts$ ./B24CS013_06_02.sh apple
Total occurrences of 'apple':6
bodike@bodike-chaithali:~/Scripts$ ./B24CS013_06_02.sh 101
Total occurrences of '101':2

```
bodike@bodike-chaithali:~/Scripts$ cat data.csv
ID,Name,Fruit,Quantity
1,Alice,Apple,10
2,Bob,Banana,5
3,Charlie,apple,7
4,David,Grape,3
5,Eve,Applepie,4
6,Frank,APPLE,8
bodike@bodike-chaithali:~/Scripts$ cat sales.csv
OrderID,Customer,Product,Quantity
101,John,Orange,12
102,Linda,apple,5
103,Mark,Banana,7
104,Susan,APPLE,10
105,Tom,AppleSauce,3
106,Emma,grape,8
107,Oliver,apple,6
bodike@bodike-chaithali:~/Scripts$
```